

Claim bank — review request

For Timur Alptunaer, MD · prepared 2026-09-15

What follows is 30 claims with their hedges, their do-not-say lists and their citations. It is long because it has to be exact. **You are being asked three questions; everything else is supporting material.**

1. Does anything in the strong tier not belong there?

Eight claims are graded **strong** — the system's grade, not yours. A strong claim is pitched with the most confidence and the least hedging, so a mis-grade there costs more than one anywhere else. Sound but over-graded → **Approve with edits**, saying what it should be. Not supportable at all → **Strike**.

2. Does anything in the preliminary tier need striking?

Seven claims rest on mechanistic, small or single-study evidence — nine participants in one case, cellular markers rather than outcomes in others. They are collected on the first page so you can **strike the entire tier in one decision** if you would rather not have early-stage findings attributed to you. That is a legitimate answer and costs little — the strong and moderate tiers carry the topics that actually get asked about.

3. Are there claims where cohort size would change your answer?

This is the one we cannot answer for you, and the reason matters.

Sample size is missing for 35 of the 47 citations. It reads *sample size not stated*, which means not stated — not inferred, not summed from subgroups, not recalled from memory. The index returns abstracts, full text was unavailable for every paper here, and only 12 citations state a figure verbatim in the abstract.

So **mark any claim where you would want the cohort size before deciding**. Those get pulled and the numbers read out of the papers first. We would rather hold a claim than have you approve one on an assumption about its size.

Journal names are absent for the same reason — dropped rather than derived from the DOI prefix, since **10.1001** covers JAMA, JAMA Internal Medicine and JAMA Cardiology alike. Every DOI resolves to the paper of record; use them where the venue matters to your judgement.

What approval commits you to

Standing attribution of these specific sentences, with their hedges attached, without you reviewing each individual pitch. Not a topic area, not a general endorsement of sauna or cold exposure. A claim not in this bank cannot be pitched, and that is enforced in code rather than by anyone remembering. Nothing has been sent and nothing can be until this is signed: the system records the bank as **awaiting_review** and refuses to run any drafting step.

Two things worth knowing before you start

Every citation was retrieved from the live literature index, not written from memory. Titles are stored exactly as returned and re-checked automatically; if a stored citation stops matching its identifier the bank fails closed.

One claim records a null result on purpose. **heat-vascular-limits-01** is a randomised trial in coronary artery disease that found no benefit, sitting alongside the positive observational findings. It is there because a source who volunteers the trial that did not work is more credible than one who does not.

Marking is per claim throughout: Approve ☐ · Approve with edits ☐ · Strike ☐. The sign-off page is at the end.

Claim bank — for review by Timur Alptunaer, MD

Prepared 2026-09-14 · 30 claims · 47 citations · version 1.0

What you are being asked to approve

These are the only claims that may be attributed to you in journalist pitches. A claim that is not on this list cannot be pitched — that is enforced in code, not by anyone remembering.

Approving this bank means: **standing attribution of these specific claims, with their hedges attached, without you reviewing each individual pitch.** You are not approving a topic area or a general endorsement of sauna and cold exposure. You are approving these sentences.

Each claim carries a **hedge** that must travel with it, and a **do-not-say** list of overclaims it must never be stretched into. If you strike a claim, it leaves the bank. If you rewrite a hedge, the new wording is what ships.

Every citation below was retrieved from the live literature index, not written from memory. Titles are stored exactly as returned and are re-checked automatically; if a stored citation stops matching its identifier, the bank fails closed rather than pitching it.

strong	8 (27%)
moderate	15 (50%)
preliminary	7 (23%)

Read this first — the 7 preliminary claims

These rest on mechanistic, small or single-study evidence. They are grouped here so you can **strike the whole tier in one decision** if you would rather not have early-stage findings attributed to you at all.

- **sleep-slowwave-01** — Passive body heating in the evening increased slow-wave sleep in a small study of older women with insomnia.
 - *Hedge*: Nine participants, one small crossover study from 1996. This is early and narrow evidence, not an established effect.
 - *Evidence*: RCT
- **heat-shock-01** — Repeated passive heat exposure raises intracellular heat shock protein 70, one of the cellular stress responses also triggered by exercise.
 - *Hedge*: This is a cellular marker, not a health outcome. Nobody has shown that raising HSP70 by heating produces a clinical benefit in humans.
 - *Evidence*: meta-analysis
- **heat-shock-02** — A single session of passive heating produces a measurable inflammatory and heat shock protein response broadly similar in direction to moderate exercise.
 - *Hedge*: Small mechanistic study measuring blood markers over hours. Similar direction is not equivalent magnitude, and no outcome was measured.
 - *Evidence*: mechanistic

- **cold-glucose-01** — In a study of men, cold-induced brown fat activity was accompanied by changes in plasma fatty acids and measures of glucose metabolism.
- *Hedge*: Small, male-only, short-duration, and measuring biochemical markers rather than diabetes risk or any clinical endpoint.
- *Evidence*: mechanistic
- **cold-affect-01** — A short cold-water immersion increased positive affect in a small imaging study, alongside changes in connectivity between brain networks.
- *Hedge*: A small first-of-its-kind imaging study. Brain-network changes are not evidence of therapeutic benefit.
- *Evidence*: mechanistic
- **heat-mood-01** — A single session of whole-body hyperthermia reduced depressive symptoms compared with a sham procedure in a randomised trial of adults with major depressive disorder.
- *Hedge*: One small trial, in a clinical population, using medically supervised whole-body hyperthermia under controlled conditions — not a domestic sauna, and not repeated dosing.
- *Evidence*: RCT
- **safety-pregnancy-02** — One review of uncomplicated pregnancy concluded sauna bathing did not affect fertility and was tolerated by healthy women, which sits alongside — not above — the hyperthermia evidence.
- *Hedge*: These two literatures genuinely disagree in emphasis. The conservative reading governs: core temperature rise in the first trimester is the risk, and this is a decision for the treating clinician.
- *Evidence*: review

Strike the tier: ☐ Keep with hedges: ☐ Decide individually below: ☐

Heat and cardiovascular outcomes

heat-cv-mortality-01 · strong · prospective cohort

In a long-running Finnish study of middle-aged men, those who used a sauna four to seven times a week had a lower rate of fatal cardiovascular events than those who went once a week.

Hedge that must travel with it. This is an observational association in a population where sauna use is lifelong and near-universal. It does not establish that sauna use caused the difference, and people who sauna often may differ in other ways.

Must never be stretched into: - “Saunas prevent heart attacks” - “Sauna use will extend your life” - “A sauna is as good as cardiovascular exercise”

Citations - Association between sauna bathing and fatal cardiovascular and all-cause mortality events. — 2015, sample size not stated (PMID 25705824, doi:10.1001/jamainternmed.2014.8187) - Sauna bathing is associated with reduced cardiovascular mortality and improves risk prediction in men and women: a prospective cohort study. — 2018, sample size not stated (PMID 30486813, doi:10.1186/s12916-018-1198-0)

Approve ☐ Approve with edits ☐ Strike ☐

heat-cv-mortality-02 · moderate · prospective cohort

The same cohort found the association held in women as well as men when it was re-examined with both sexes included.

Hedge that must travel with it. One cohort, one country, one bathing culture. It has not been replicated in a population where sauna use is occasional rather than habitual.

Must never be stretched into: - “Saunas protect women from heart disease” - “This applies to any population”

Citations - Sauna bathing is associated with reduced cardiovascular mortality and improves risk prediction in men and women: a prospective cohort study. — 2018, sample size not stated (PMID 30486813, doi:10.1186/s12916-018-1198-0)

Approve ☐ Approve with edits ☐ Strike ☐

heat-bp-01 · moderate · mechanistic

A single sauna session lowers blood pressure for a period afterwards, with systolic pressure falling by roughly seven points in one controlled study.

Hedge that must travel with it. This is a short-term effect measured after one session, not evidence of lasting blood-pressure control.

Must never be stretched into: - “Sauna treats hypertension” - “You can reduce or stop blood-pressure medication” - “Sauna replaces antihypertensives”

Citations - Sauna exposure leads to improved arterial compliance: Findings from a non-randomised experimental study. — 2018, sample size not stated (PMID 29048215, doi:10.1177/2047487317737629)

Approve ☐ Approve with edits ☐ Strike ☐

heat-vascular-01 · moderate · RCT

Eight weeks of regular passive heat exposure improved blood-vessel function and arterial stiffness in sedentary adults in a controlled trial.

Hedge that must travel with it. A small trial in young sedentary adults using hot-water immersion, not sauna, and measuring intermediate markers rather than clinical events.

Must never be stretched into: - “Heat therapy reverses cardiovascular disease” - “Sauna unclogs arteries”

Citations - Passive heat therapy improves endothelial function, arterial stiffness and blood pressure in sedentary humans. — 2016, sample size not stated (PMID 27270841, doi:10.1113/JP272453)

Approve ☐ Approve with edits ☐ Strike ☐

heat-vascular-limits-01 · moderate · RCT

When sauna bathing was tested as an eight-week randomised intervention in adults who already had coronary artery disease, it did not produce the vascular improvements seen in healthy sedentary people.

Hedge that must travel with it. A null result in an established-disease population. It does not negate the observational mortality findings, but it does show the benefit is not automatic in patients.

Must never be stretched into: - “Sauna is proven therapy for heart patients” - “Heart patients should use sauna to treat their disease”

Citations - Finnish sauna bathing and vascular health of adults with coronary artery disease: a randomized controlled trial. — 2023, sample size not stated (PMID 37650138, doi:10.1152/jappphysiol.00322.2023) - Passive heat therapy improves endothelial function, arterial stiffness and blood pressure in sedentary humans. — 2016, sample size not stated (PMID 27270841, doi:10.1113/JP272453)

Approve ☐ Approve with edits ☐ Strike ☐

safety-stable-cad-01 · moderate · review

For most people with stable coronary heart disease, sauna bathing is generally well tolerated.

Hedge that must travel with it. Well tolerated is not the same as beneficial — the randomised trial in coronary artery disease found no vascular improvement. Individual clearance still comes from the treating clinician.

Must never be stretched into: - “Sauna is safe for everyone with heart disease” - “Sauna treats coronary disease” - “Heart patients should sauna for their condition”

Citations - Benefits and risks of sauna bathing. — 2001, sample size not stated (PMID 11165553, doi:10.1016/s0002-9343(00)00671-9) - Finnish sauna bathing and vascular health of adults with coronary artery disease: a randomized controlled trial. — 2023, sample size not stated (PMID 37650138, doi:10.1152/jappphysiol.00322.2023)

Approve ☐ Approve with edits ☐ Strike ☐

Heat and mortality

heat-allcause-01 · moderate · prospective cohort

Frequent sauna bathing was associated with lower all-cause mortality, and the association was present across levels of cardiorespiratory fitness.

Hedge that must travel with it. Observational. Fitness was measured at baseline only, and unmeasured differences between frequent and infrequent bathers cannot be ruled out.

Must never be stretched into: - “Sauna use adds years to your life” - “Sauna can substitute for being fit”

Citations - Joint associations of sauna bathing and cardiorespiratory fitness on cardiovascular and all-cause mortality risk: a long-term prospective cohort study. — 2018, sample size not stated (PMID 28972808, doi:10.1080/07853890.2017.1387927) - Association between sauna bathing and fatal cardiovascular and all-cause mortality events. — 2015, sample size not stated (PMID 25705824, doi:10.1001/jamainternmed.2014.8187)

Approve ☐ Approve with edits ☐ Strike ☐

Heat and cognition

heat-cognition-01 · moderate · prospective cohort

In the same Finnish cohort, more frequent sauna bathing was associated with a lower recorded rate of dementia and Alzheimer's disease.

Hedge that must travel with it. Observational, men only, and dementia diagnosis in a cohort of this era is imprecise. Reverse causation is plausible: early cognitive decline may reduce sauna use rather than the reverse.

Must never be stretched into: - “Saunas prevent dementia” - “Sauna protects the brain” - “Sauna reduces Alzheimer's risk”

Citations - Sauna bathing is inversely associated with dementia and Alzheimer's disease in middle-aged Finnish men. — 2017, sample size not stated (PMID 27932366, doi:10.1093/ageing/afw212)

Approve ☐ Approve with edits ☐ Strike ☐

Heat and sleep

sleep-passive-heating-01 · moderate · meta-analysis

Warming the body one to two hours before bed shortens the time it takes to fall asleep, by about ten minutes on average across the pooled trials.

Hedge that must travel with it. The pooled evidence is for warm baths and showers, not saunas, and timing matters — the effect depends on heating well before bed rather than immediately before.

Must never be stretched into: - “Sauna cures insomnia” - “A sauna before bed guarantees better sleep” - “Heat is a treatment for sleep disorders”

Citations - Before-bedtime passive body heating by warm shower or bath to improve sleep: A systematic review and meta-analysis. — 2019, sample size not stated (PMID 31102877, doi:10.1016/j.smrv.2019.04.008)

Approve ☐ Approve with edits ☐ Strike ☐

sleep-slowwave-01 · preliminary · RCT

Passive body heating in the evening increased slow-wave sleep in a small study of older women with insomnia.

Hedge that must travel with it. Nine participants, one small crossover study from 1996. This is early and narrow evidence, not an established effect.

Must never be stretched into: - “Heat increases deep sleep” - “Sauna improves sleep architecture” - “Proven to increase slow-wave sleep”

Citations - Effects of passive body heating on the sleep of older female insomniacs. — 1996, n=9 (PMID 8736588, doi:10.1177/089198879600900203) - Before-bedtime passive body heating by warm shower or bath to improve

sleep: A systematic review and meta-analysis. — 2019, sample size not stated (PMID 31102877, doi:10.1016/j.smr.2019.04.008)

Approve ☐ Approve with edits ☐ Strike ☐

Heat shock response

heat-shock-01 · preliminary · meta-analysis

Repeated passive heat exposure raises intracellular heat shock protein 70, one of the cellular stress responses also triggered by exercise.

Hedge that must travel with it. This is a cellular marker, not a health outcome. Nobody has shown that raising HSP70 by heating produces a clinical benefit in humans.

Must never be stretched into: - “Sauna triggers cellular repair that prevents disease” - “Heat shock proteins are why sauna extends lifespan” - “Sauna gives you the benefits of exercise”

Citations - Heat acclimation-induced intracellular HSP70 in humans: a meta-analysis. — 2020, sample size not stated (PMID 31823288, doi:10.1007/s12192-019-01059-y)

Approve ☐ Approve with edits ☐ Strike ☐

heat-shock-02 · preliminary · mechanistic

A single session of passive heating produces a measurable inflammatory and heat shock protein response broadly similar in direction to moderate exercise.

Hedge that must travel with it. Small mechanistic study measuring blood markers over hours. Similar direction is not equivalent magnitude, and no outcome was measured.

Must never be stretched into: - “A sauna session equals a workout” - “Sauna is exercise for people who cannot exercise” - “Passive heating replaces physical activity”

Citations - The effect of passive heating on heat shock protein 70 and interleukin-6: A possible treatment tool for metabolic diseases? — 2017, sample size not stated (PMID 28944271, doi:10.1080/23328940.2017.1288688)

Approve ☐ Approve with edits ☐ Strike ☐

Heat and mood

cold-mood-01 · moderate · systematic review

Across the pooled trials, cold-water immersion is followed by a short-lived improvement in self-reported mood and wellbeing.

Hedge that must travel with it. Effects in the pooled data are short-lived, measured mostly in healthy volunteers, and self-reported in studies that cannot be blinded — people know they got in cold water.

Must never be stretched into: - “Cold plunging treats depression” - “Cold water is an antidepressant” - “Cold exposure fixes anxiety”

Citations - Effects of cold-water immersion on health and wellbeing: A systematic review and meta-analysis. — 2025, sample size not stated (PMID 39879231, doi:10.1371/journal.pone.0317615)

Approve ☐ Approve with edits ☐ Strike ☐

cold-affect-01 · preliminary · mechanistic

A short cold-water immersion increased positive affect in a small imaging study, alongside changes in connectivity between brain networks.

Hedge that must travel with it. A small first-of-its-kind imaging study. Brain-network changes are not evidence of therapeutic benefit.

Must never be stretched into: - “Cold plunging rewires your brain” - “Cold exposure changes brain function for the better” - “Neuroscience proves cold plunging works”

Citations - Short-Term Head-Out Whole-Body Cold-Water Immersion Facilitates Positive Affect and Increases Interaction between Large-Scale Brain Networks. — 2023, sample size not stated (PMID 36829490, doi:10.3390/biology12020211) - Effects of cold-water immersion on health and wellbeing: A systematic review and meta-analysis. — 2025, sample size not stated (PMID 39879231, doi:10.1371/journal.pone.0317615)

Approve ☐ Approve with edits ☐ Strike ☐

heat-mood-01 · preliminary · RCT

A single session of whole-body hyperthermia reduced depressive symptoms compared with a sham procedure in a randomised trial of adults with major depressive disorder.

Hedge that must travel with it. One small trial, in a clinical population, using medically supervised whole-body hyperthermia under controlled conditions — not a domestic sauna, and not repeated dosing.

Must never be stretched into: - “Saunas treat depression” - “Sauna is an antidepressant” - “Heat therapy is proven for mental health” - “Use a sauna instead of medication”

Citations - Whole-Body Hyperthermia for the Treatment of Major Depressive Disorder: A Randomized Clinical Trial. — 2016, sample size not stated (PMID 27172277, doi:10.1001/jamapsychiatry.2016.1031)

Approve ☐ Approve with edits ☐ Strike ☐

Cold and metabolic markers

cold-bat-01 · moderate · meta-analysis

Cold exposure activates brown adipose tissue in adults, which increases energy expenditure while the cold lasts.

Hedge that must travel with it. Brown fat activation is measured in laboratory conditions over short exposures. The energy involved is small and has not been shown to produce meaningful weight change.

Must never be stretched into: - “Cold plunging burns fat” - “Cold exposure is a weight-loss tool” - “Brown fat activation will make you leaner”

Citations - Metabolic Effects of Brown Adipose Tissue Activity Due to Cold Exposure in Humans: A Systematic Review and Meta-Analysis of RCTs and Non-RCTs. — 2024, sample size not stated (PMID 38540150, doi:10.3390/biomedicines12030537)

Approve ☐ Approve with edits ☐ Strike ☐

cold-glucose-01 · preliminary · mechanistic

In a study of men, cold-induced brown fat activity was accompanied by changes in plasma fatty acids and measures of glucose metabolism.

Hedge that must travel with it. Small, male-only, short-duration, and measuring biochemical markers rather than diabetes risk or any clinical endpoint.

Must never be stretched into: - “Cold plunges improve insulin sensitivity” - “Cold exposure helps manage blood sugar” - “Cold plunging prevents diabetes”

Citations - Cold-Induced Brown Adipose Tissue Activity Alters Plasma Fatty Acids and Improves Glucose Metabolism in Men. — 2017, sample size not stated (PMID 28945846, doi:10.1210/jc.2017-01250) - Metabolic Effects of Brown Adipose Tissue Activity Due to Cold Exposure in Humans: A Systematic Review and Meta-Analysis of RCTs and Non-RCTs. — 2024, sample size not stated (PMID 38540150, doi:10.3390/biomedicines12030537)

Approve ☐ Approve with edits ☐ Strike ☐

Cold, recovery and its limits

cold-recovery-01 · strong · meta-analysis

Cold-water immersion after hard exercise reduces muscle soreness in the days that follow.

Hedge that must travel with it. The reliable effect is on perceived soreness. Effects on actual performance recovery are smaller and less consistent across studies.

Must never be stretched into: - “Cold plunging speeds healing” - “Cold immersion repairs muscle damage” - “Ice baths make you recover faster”

Citations - Effects of Cold Water Immersion and Contrast Water Therapy for Recovery From Team Sport: A Systematic Review and Meta-analysis. — 2017, sample size not stated (PMID 27398915, doi:10.1519/JSC.0000000000001559) - Contrast water therapy and exercise induced muscle damage: a systematic review and meta-analysis. — 2013, sample size not stated (PMID 23626806, doi:10.1371/journal.pone.0062356)

Approve ☐ Approve with edits ☐ Strike ☐

cold-hypertrophy-limits-01 · strong · RCT

Regular cold-water immersion straight after strength training blunts muscle growth compared with passive recovery.

Hedge that must travel with it. This is the trade-off that matters for anyone lifting weights: the same cold that reduces soreness also attenuates the adaptation. Strength gains appear less affected than muscle size.

Must never be stretched into: - “Cold plunging is bad for you” - “Never use cold water after exercise” - “Cold immersion causes muscle loss”

Citations - Post-exercise cold water immersion attenuates acute anabolic signalling and long-term adaptations in muscle to strength training. — 2015, n=21 (PMID 26174323, doi:10.1113/JP270570) - Cold water immersion attenuates anabolic signaling and skeletal muscle fiber hypertrophy, but not strength gain, following whole-body resistance training. — 2019, n=16 (PMID 31513450, doi:10.1152/jappphysiol.00127.2019)

Approve ☐ Approve with edits ☐ Strike ☐

cold-timing-01 · moderate · RCT

Because cold immersion interferes with the muscle-building response, timing it away from strength sessions is the practical way to keep the soreness benefit without the trade-off.

Hedge that must travel with it. A reasonable inference from the trial evidence on timing and adaptation, not a protocol that has itself been tested head to head.

Must never be stretched into: - “The optimal cold plunge protocol is proven” - “Science says plunge X hours after training”

Citations - Cold water immersion attenuates anabolic signaling and skeletal muscle fiber hypertrophy, but not strength gain, following whole-body resistance training. — 2019, n=16 (PMID 31513450, doi:10.1152/jappphysiol.00127.2019) - Post-exercise cold water immersion attenuates acute anabolic signalling and long-term adaptations in muscle to strength training. — 2015, n=21 (PMID 26174323, doi:10.1113/JP270570)

Approve ☐ Approve with edits ☐ Strike ☐

contrast-therapy-01 · moderate · meta-analysis

Alternating hot and cold water after exercise reduces muscle soreness, performing comparably to cold-water immersion alone in pooled trials.

Hedge that must travel with it. Pooled from small studies with varied protocols. There is no established optimal ratio of hot to cold, and contrast therapy has not been shown to beat cold alone.

Must never be stretched into: - “Contrast therapy is the best recovery method” - “Hot-cold cycling flushes toxins” - “Contrast therapy boosts circulation to heal injury”

Citations - Contrast water therapy and exercise induced muscle damage: a systematic review and meta-analysis. — 2013, sample size not stated (PMID 23626806, doi:10.1371/journal.pone.0062356) - Effects of Cold Water Immersion and Contrast Water Therapy for Recovery From Team Sport: A Systematic Review and Meta-analysis. — 2017, sample size not stated (PMID 27398915, doi:10.1519/JSC.0000000000001559)

Approve ☐ Approve with edits ☐ Strike ☐

Hydration and body mass

hydration-01 · strong · mechanistic

Sweat loss in a sauna typically runs around 0.6 to 1 kilogram per hour, which is why fluid replacement matters more than most people assume.

Hedge that must travel with it. Rates vary substantially with body size, heat tolerance and session length. These are typical figures, not a prescription.

Must never be stretched into: - “Sauna detoxifies the body through sweat” - “Sweating removes heavy metals or toxins”

Citations - Human thermoregulation in sauna. — 1988, sample size not stated (PMID 3218894) - Sauna-induced body mass loss in young sedentary women and men. — 2014, n=674 (PMID 25614882, doi:10.1155/2014/307421)

Approve ☐ Approve with edits ☐ Strike ☐

hydration-02 · strong · mechanistic

Weight lost during a sauna session is fluid, and it returns as soon as you rehydrate.

Hedge that must travel with it. Body-mass loss in a sauna is water, not fat. Treating it as fat loss is the most common misreading of the scale after a session.

Must never be stretched into: - “Saunas burn fat” - “You can lose weight in a sauna” - “Sauna is a weight-loss tool” - “A sauna session burns X calories of fat”

Citations - Sauna-induced body mass loss in young sedentary women and men. — 2014, n=674 (PMID 25614882, doi:10.1155/2014/307421) - Human thermoregulation in sauna. — 1988, sample size not stated (PMID 3218894)

Approve ☐ Approve with edits ☐ Strike ☐

Safety, contraindications and who should not

safety-contraindications-01 · strong · review

Sauna bathing is not advised for people with unstable angina, a recent heart attack, or severe aortic stenosis.

Hedge that must travel with it. These are the established contraindications. Anyone with a cardiac diagnosis should ask their own physician rather than rely on a general rule.

Must never be stretched into: - “Sauna is dangerous for heart patients generally” - “People with heart disease must avoid saunas”

Citations - Benefits and risks of sauna bathing. — 2001, sample size not stated (PMID 11165553, doi:10.1016/s0002-9343(00)00671-9)

Approve ☐ Approve with edits ☐ Strike ☐

safety-alcohol-01 · strong · case series

Alcohol and sauna are a genuinely dangerous combination, and alcohol is implicated in a large share of sauna-related deaths.

Hedge that must travel with it. In the Swedish forensic series, alcohol was involved in roughly 44 per cent of the deaths examined. Bathing alone compounds the risk.

Must never be stretched into: - “Sauna is dangerous” - “A drink before the sauna is fine in moderation”

Citations - Sauna deaths in Sweden, 1992-2003. — 2008, n=77 (PMID 19749613, doi:10.1097/PAF.0b013e318145ae05) - Alcohol and sauna bathing: effects on cardiac rhythm, blood pressure, and serum electrolyte and cortisol concentrations. — 1992, n=10 (PMID 1588256, doi:10.1111/j.1365-2796.1992.tb00941.x)

Approve ☐ Approve with edits ☐ Strike ☐

safety-arrhythmia-01 · moderate · mechanistic

Combining alcohol with sauna heat raises the risk of cardiac rhythm disturbance, with the hangover phase carrying particular risk.

Hedge that must travel with it. Demonstrated in a small controlled study of ten healthy men plus forensic case data, not a large clinical trial.

Must never be stretched into: - “Sauna causes arrhythmia” - “Sauna is unsafe for anyone with palpitations”

Citations - Alcohol and sauna bathing: effects on cardiac rhythm, blood pressure, and serum electrolyte and cortisol concentrations. — 1992, n=10 (PMID 1588256, doi:10.1111/j.1365-2796.1992.tb00941.x) - Sauna deaths in Sweden, 1992-2003. — 2008, n=77 (PMID 19749613, doi:10.1097/PAF.0b013e318145ae05)

Approve ☐ Approve with edits ☐ Strike ☐

safety-alone-01 · moderate · case series

Bathing alone is a recurring feature of fatal sauna incidents, which is a practical argument for not using a home sauna alone after drinking.

Hedge that must travel with it. Drawn from a forensic case series, which describes deaths that occurred and cannot give a rate of risk for ordinary use.

Must never be stretched into: - “Never use a sauna alone” - “Home saunas are dangerous”

Citations - Sauna deaths in Sweden, 1992-2003. — 2008, n=77 (PMID 19749613, doi:10.1097/PAF.0b013e318145ae05)

Approve ☐ Approve with edits ☐ Strike ☐

safety-pregnancy-01 · strong · review

Raising core body temperature in early pregnancy is associated with an increased risk of birth defects, so pregnancy is the clearest case for medical advice before heat exposure.

Hedge that must travel with it. The evidence concerns sustained elevation of maternal core temperature during the first trimester. Pregnant women should be directed to their own clinician, not to a general guideline.

Must never be stretched into: - “Saunas cause birth defects” - “Pregnant women must never use a sauna” - “Sauna is safe during pregnancy”

Citations - Update on the gestational effects of maternal hyperthermia. — 2020, sample size not stated (PMID 32686349, doi:10.1002/bdr2.1696)

Approve ☐ Approve with edits ☐ Strike ☐

safety-pregnancy-02 · preliminary · review

One review of uncomplicated pregnancy concluded sauna bathing did not affect fertility and was tolerated by healthy women, which sits alongside — not above — the hyperthermia evidence.

Hedge that must travel with it. These two literatures genuinely disagree in emphasis. The conservative reading governs: core temperature rise in the first trimester is the risk, and this is a decision for the treating clinician.

Must never be stretched into: - “Sauna is safe in pregnancy” - “Research shows pregnant women can sauna normally” - “Sauna does not affect pregnancy”

Citations - Benefits and risks of sauna bathing. — 2001, sample size not stated (PMID 11165553, doi:10.1016/s0002-9343(00)00671-9) - Update on the gestational effects of maternal hyperthermia. — 2020, sample size not stated (PMID 32686349, doi:10.1002/bdr2.1696)

Approve ☐ Approve with edits ☐ Strike ☐

Sauna bathing is generally tolerated by healthy children in the cultures where it is customary, but children heat up faster than adults and need shorter sessions and supervision.

Hedge that must travel with it. Tolerance in a sauna-bathing culture with customary practice is not a green light for unsupervised use, and thermoregulation differs by body size.

Must never be stretched into: - “Saunas are safe for kids” - “Children can use a sauna like adults”

Citations - Benefits and risks of sauna bathing. — 2001, sample size not stated (PMID 11165553, doi:10.1016/s0002-9343(00)00671-9) - Human thermoregulation in sauna. — 1988, sample size not stated (PMID 3218894)

Approve ☐ Approve with edits ☐ Strike ☐

Gaps — deliberately not filled

These were searched and left empty rather than supported with weaker evidence relabelled as strong:

- **Sauna and sleep specifically.** The usable pooled evidence is for warm baths and showers before bed, not saunas. The sleep claims say so.
- **Cold exposure and immune function.** Popular topic, no human evidence found at a standard worth attaching a physician's name to.
- **Sauna and detoxification.** Supportable only from marketing material. Left out entirely, and named in the do-not-say lists.
- **Product comparisons.** No head-to-head trials exist between the equipment categories the store sells, so no comparative claim is offered.

What this document cannot tell you

Sample size is missing for most citations, and journal name for all of them. This is a limitation of the source, not an oversight.

The research index returns title, authors, identifiers, dates and abstract. It returns no journal field, and full text was unavailable for every paper in this bank, so sample sizes could not be read out of article bodies. The 12 citations showing a count are those where the figure appears verbatim in the retrieved abstract. The rest read *sample size not stated* — that means not stated, not zero, and it was not inferred, summed from subgroups, or recalled from memory.

Journal name was deliberately dropped rather than derived from the DOI prefix: prefix 10.1001 covers JAMA, JAMA Internal Medicine and JAMA Cardiology alike, so deriving it risks naming a journal a paper was never published in. Every DOI below resolves to the paper of record — please use it where the venue matters to your judgement.

Where the evidence genuinely disagrees

safety-pregnancy-01 and safety-pregnancy-02 point in different directions, and both are in the bank on purpose. The conservative reading governs in both hedges. If you would rather drop the second entirely, striking it loses nothing the first does not already cover.

heat-vascular-limits-01 records a **null** randomised result in coronary artery disease alongside the positive observational findings. It is included because a source who volunteers the trial that did not work is more credible than one who does not.

Sign-off

I approve the claims marked above for standing attribution to me in media pitches, with their hedges intact, without per-pitch review. Claims I have struck are removed from the bank and cannot be pitched.

Name	Date	Signature

Re-approval is required if any claim text or hedge changes. The bank records approval status; until it is signed it reads `awaiting_review`.